

Robust Branch-Safe Participation for Sequential Mobile Crowdsensing: Model Uncertainty and Disclosure Floors

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Abstract

Keeping coordinates local does not prevent location leakage from regional task outcomes. A truthful report implies presence in the task region, while silence can also update an observer’s belief. We study this linkable binary channel under uncertain delivery, willingness, and timely completion. Branch-safe participation (BSP) chooses the maximal scalar gate that bounds posterior peak after both outputs under a declared model, while robust BSP (R-BSP) intersects feasible gates across a finite ambiguity set. We further show that a truthful positive report has an unavoidable posterior peak that is independent of any positive thinning probability. Disclosure-floor-aware BSP (DF-BSP) therefore introduces an explicit report-side threshold $\tau \geq \rho$ while retaining the original BSP cap for silence; RDF-BSP combines this asymmetric contract with finite-model robustness. In a post-hoc extension, model-input-group isolation leaves 23 GeoLife development windows and 12 validation windows for model-set construction, while 53 isolated test windows remain excluded from that construction. Against a stronger population-prior replay attacker, strict data-driven R-BSP records 0.2% observed local-cap violations in this replay but retains only 18.8% of opportunities. At $\tau = 0.5$, RDF-BSP retains 23.5%

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of 1–8-cell opportunities and 26.0% inverse-area-weighted utility, with a 2.00% branch-specific cap-violation rate and a 14.7% violation rate of the original BSP local cap. Additional synthetic walks with new seeds show the same qualitative privacy–service trade-off. The guarantees remain channel- and model-set-specific and do not cover arbitrary auxiliary knowledge.

Keywords: Mobile crowdsensing, location privacy, Bayesian inference, robust participation, disclosure floor, adaptive tasks

1. Introduction

Mobile crowdsensing (MCS) allows personal devices to contribute observations for public regions and time windows while keeping exact coordinates locally. Local storage, however, does not remove the information carried by participation outcomes. If a linkable account successfully completes a task valid only inside a region, that completion is evidence of presence there; repeated outcomes can support sequential location inference. Silence is informative as well because delivery, willingness, deadline success, and privacy filtering all affect its probability.

Task-outcome inference is not new. TMarkov studies inference from task acceptance/refusal and mobility correlations [1], while recent fingerprint-based assignment also protects worker and task locations [2]. Our narrower question is different: when the platform deliberately observes a truthful binary completion channel—a valid in-region report or silence—can the client regulate the location information in both branches without false reports or a hidden task-selection protocol?

We model that visible channel and develop branch-safe participation (BSP), a public-history scalar gate that maximizes participation subject to a bound on posterior peak after every positive-probability output. The rule is model conditional: it is neither differential privacy nor a guarantee against arbitrary auxiliary knowledge.

We then address two limitations of a single-model scalar gate. Robust BSP (R-BSP) intersects the feasible gates of a finite ambiguity set; a development/validation procedure constructs a replay model set instead of relying only on a hand-selected stress grid. We also show that truthful successful reports have an unavoidable disclosure floor that scalar thinning cannot reduce. Disclosure-floor-aware BSP (DF-BSP) therefore introduces a separate report-side threshold τ , while keeping silence at the original BSP cap; RDF-BSP combines this asymmetric contract with finite-model robustness.

The paper makes four contributions. First, it specifies the report/silence observation boundary and derives a maximal scalar gate with a conditional sequential bound. Second, it derives the maximal common gate for finite model sets and evaluates model-set failures rather than assuming a correct attacker model. Third, it formalizes the positive-report disclosure floor and an asymmetric privacy–service contract for fine-region tasks. Fourth, it evaluates these mechanisms on synthetic mobility and disjoint GeoLife development/validation/test users, including informed-attacker stress, boundary cases, size-aware utility, and task-channel adaptations of PRIVIC [3] and pointwise maximal leakage (PML) [4]. Negative results—including severe utility loss under strict robust caps—are retained in the main analysis.

2. Related Work

MCS privacy mechanisms mostly protect information before or during assignment. Bilateral and task-location schemes protect worker/requester or task coordinates [5, 6]. Other work uses differential obfuscation [7], secret-sharing recruitment [8], encrypted location/reward optimization [9], and local differential privacy or geo-indistinguishability [10]. Recent approaches additionally consider temporal–spatial differential privacy [11], mobility-prediction-aware perturba-

tion [12], encrypted quality-aware allocation [13], federated local task selection [14], and blockchain/ZKP protocols [15]. These mechanisms protect coordinate, matching, recruitment, or transaction interfaces. Their guarantees and reported overhead cannot be transferred directly to a visible truthful report/silence channel.

HECTA is a close recent precedent because it addresses inference from task-selection sequences [16]. It masks genuine choices among dummy tasks using cryptographic processing and dummy-task mixing under a non-collusion assumption between the service provider and its auxiliary party. BSP instead leaves task regions and linkable truthful completion outcomes visible and regulates the Bayesian posterior induced by those outputs. The trust model, observed transcript, and privacy metric are therefore different; we do not rank the two guarantees or reproduce HECTA numerically.

Bayesian privacy measures prior-to-posterior change [17]. Adaptive differential privacy concerns a different formal guarantee [18], while PRIVIC protects randomized location releases under geo-indistinguishability [3]. BSP instead limits posterior concentration under an explicit mobility and availability model. Because a truthful successful report has zero support outside its task region, BSP is not a differentially private coordinate-release mechanism. PML provides a useful optimization-based channel comparator [4]. Complete comparator derivations and implementation checks are provided in the Supplementary Material to keep the main text focused.

3. Proposed Branch-Safe Participation Framework

Figure 1 separates private sensing state from the information available to the platform. Public task metadata enters both the eligibility check and the participation filter. The same observable outcome updates the client’s declared-model belief and

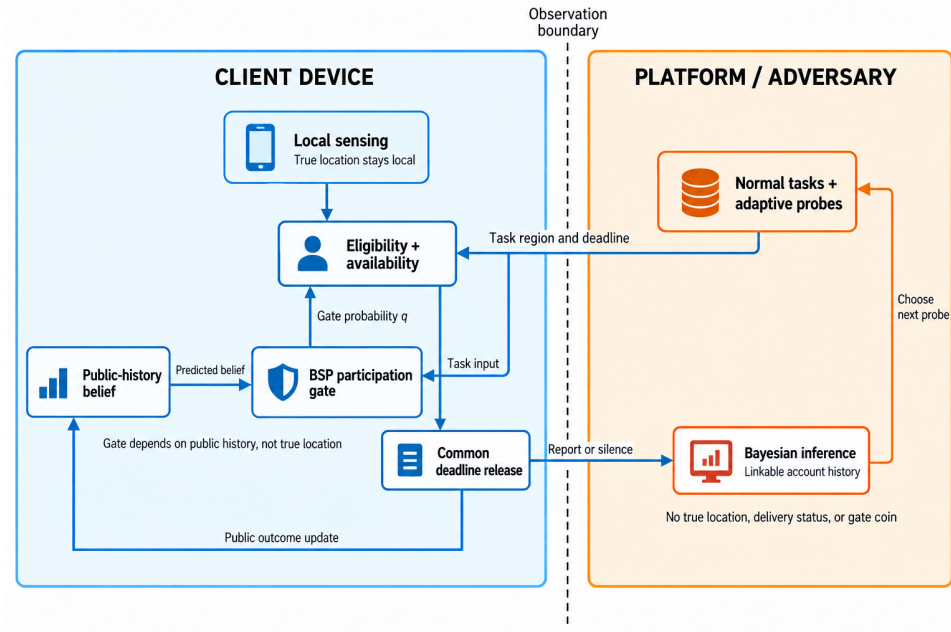


Figure 1: System architecture and observation boundary. A linkable account produces a truthful report or silence at a common deadline. Normal tasks are exogenous, while designated adaptive probes are selected from public history. The BSP gate uses predicted belief and task metadata; true location is used locally for eligibility. The platform does not directly observe true location, task-delivery status, or the gate coin. This is a conceptual schematic of the stated system model, not an experimental result.

the attacker’s inference.

3.1. System and Observation Model

Let $X_t \in \mathcal{X}$ denote a user’s location cell at the sampling instant in slot t , where $|\mathcal{X}| = N$. A task specifies a public region $S_t \subseteq \mathcal{X}$, a sampling slot, and a deadline. A valid report requires an observation collected inside S_t at that sampling instant and submitted by the deadline. Receiving or accepting a task is not sufficient. The simulator exposes each task outcome only at its common slot deadline; it does not reveal within-slot submission timing.

Let D_t, W_t, L_t denote successful delivery, willingness to participate, and timely

completion. In the baseline model these are independent Bernoulli variables, independent of location conditional on public history, with product probability α_t . The filter additionally samples $G_t \sim \text{Bernoulli}(q_t)$ independently. The only published outcome is

$$Y_t = \mathbf{1}\{X_t \in S_t\}D_tW_tL_tG_t \in \{0, 1\}. \quad (1)$$

Here $Y_t = 0$ means that no valid report appeared by the deadline. There are no separate acknowledgments of delivery, online status, refusal, or filter rejection. Reports do not disclose additional measurement content in this model. If such content, network metadata, or fine-grained submission times are visible, the observation channel must be extended before applying the guarantee.

This design represents opportunistic sampling at a specified instant, not continuous movement toward a newly assigned destination. Users do not change their paths in response to tasks. The simulated normal task stream is public and exogenous. Probe tasks share the same completion requirements as normal tasks; the evaluator’s normal/probe label is not an input to the client’s gate.

3.2. Privacy Objective and Bayesian Observation Channel

Let $b_t(i) = \Pr(X_t = i \mid H_{t-1}, S_t)$ be the predicted belief before the current output, and abbreviate $b = b_t$, $h_i = \alpha_t \mathbf{1}\{i \in S_t\}$, and $m = \sum_i b_i h_i$. A gate is a deterministic function of public history and the current task; its random coin is private. For $q > 0$ and $m > 0$, the report posterior is

$$b_i^+ = \frac{b_i h_i}{m}. \quad (2)$$

For a positive-probability silence branch, it is

$$b_i^- = \frac{b_i(1 - qh_i)}{1 - qm}. \quad (3)$$

Zero-probability branches impose no condition. For example, when $q = 0$, silence leaves the predicted belief unchanged. With $\alpha_t < 1$, silence generally preserves

positive probability inside S_t . Between slots, a public transition matrix predicts the next location belief.

We evaluate posterior concentration through $\|b\|_\infty$, as well as attack hit rate, error, log loss, and calibration. A count of plausible cells alone is not a probability bound. A posterior with many nonzero entries can still place nearly all probability on one cell.

3.3. Branch-Safe Participation

Fix a desired posterior threshold ρ and set $c = \max\{\rho, \|b\|_\infty\}$. The local objective is to maximize $q \in [0, 1]$ subject to $\|b^y\|_\infty \leq c$ for each positive-probability output y . Using c prevents a request from being infeasible solely because the predicted belief already exceeds ρ . In that situation the rule prevents additional peak concentration relative to the current belief; it cannot restore a previously violated absolute threshold.

If the report posterior in Eq. (2) already exceeds c , no positive scalar gate can repair that branch and BSP sets $q = 0$. Otherwise, writing $d_i = cm - b_i h_i$, the gate is

$$q^* = \min \left\{ 1, \min_{i: d_i > 0} \frac{c - b_i}{d_i} \right\}, \quad (4)$$

where an empty inner minimum is $+\infty$; when $m = 0$, the output is uninformative and $q^* = 1$. Section 4 proves that this is the maximal feasible scalar gate.

The calculation takes $O(N)$ time for one task and stores an N -cell belief. Maximizing q maximizes the current expected completion probability qm within this scalar family. It does not optimize the utility of future tasks. In particular, consuming available posterior concentration now can restrict later participation. A nonmyopic scheduler or a mechanism whose gate depends on private location would be a different optimization problem.

3.4. Robust BSP under Model Uncertainty

We model uncertainty through a finite public ambiguity set; no distribution-free guarantee is claimed. Let $\mathcal{M} = \{M_1, \dots, M_K\}$ be fixed independently of the current private state. Model M_k maintains its own public-history belief $b_t^{(k)}$, transition rule, and availability probability $\alpha_t^{(k)}$. For the current task, let q_k^* be the BSP gate from Eq. (4) under that model and let $c_k = \max\{\rho, \|b_t^{(k)}\|_\infty\}$. Robust BSP (R-BSP) uses

$$q_R^* = \min_{1 \leq k \leq K} q_k^*. \quad (5)$$

All member beliefs are updated from the same public task and report/silence history; none receives the true location or the gate coin.

R-BSP costs $O(KN)$ per task when each member uses the N -cell implementation above. Enlarging $\mathcal{M}$ can only weakly decrease the feasible gate, so model-set construction is itself a privacy–utility design choice. The formal finite-model guarantee and its out-of-set limitation are analyzed in Section 4.

3.5. Disclosure-Floor-Aware BSP

Strict BSP can make fine-region service impossible for a reason that scalar thinning cannot repair. For predicted belief b and region S , define the truthful positive-report disclosure floor

$$\phi(b, S) = \frac{\max_{i \in S} b_i}{\sum_{j \in S} b_j}, \quad (6)$$

when the denominator is positive. The common availability factor α and any positive scalar gate cancel from this conditional posterior. Hence every truthful successful report from S has posterior peak $\phi(b, S)$, regardless of how small a nonzero q is. Under a uniform prior and an m -cell region, $\phi = 1/m$; a strict $\rho = 0.1$ rule therefore cannot release a truthful success for fewer than ten cells.

To express rather than conceal this service limitation, let $\tau \in [\rho, 1]$ be a declared positive-report threshold. DF-BSP uses asymmetric branch caps

$$c_{\text{DF}}^+ = \max\{\tau, \|b\|_\infty\}, \quad c_{\text{DF}}^- = \max\{\rho, \|b\|_\infty\}. \quad (7)$$

It rejects the task when $\phi(b, S) > c_{\text{DF}}^+$. Otherwise, the positive branch is admissible by construction and the gate is maximized subject to the original BSP silence constraint $\|b^-\|_\infty \leq c_{\text{DF}}^-$. Thus the structural relaxation applies only to truthful reports; silence is not allowed to inherit the relaxed report-side threshold.

Robust DF-BSP (RDF-BSP) applies this asymmetric construction independently to every $M_k \in \mathcal{M}$ and takes the minimum member-wise gate. It is therefore the maximal common scalar gate satisfying all member-specific report ceilings and strict silence caps. Importantly, $\tau > \rho$ relaxes only the truthful-report branch; it is not evidence that the original ρ cap is preserved after reports. We consequently report both violation of the original BSP local cap $\max\{\rho, \|b\|_\infty\}$ and violation of the declared DF branch-specific caps.

Operationally, the client predicts its public-history belief, computes the applicable BSP/R-BSP/DF/RDF gate from public task metadata, samples the private gate coin locally, and releases only a valid report at the common deadline. The observer updates from the same report or silence. The next section establishes the formal guarantees and their scope.

4. Privacy Guarantees and Robustness Analysis

The mechanisms above are Bayesian output-channel controls rather than cryptographic protocols. Their security argument is therefore stated in terms of posterior branches, sequential model assumptions, model mismatch, and the unavoidable information in a truthful positive report.

4.1. Threat Model and Attack Surface

The platform links outcomes to an account, knows the task regions, deadlines, filtering algorithm, and its parameters, and retains the account’s public history. At designated probe slots, it may choose a task region from a finite public library using that history. It does not observe the simulator’s actual location, future trajectory, delivery draws, willingness, or deadline-success draws. A prior-only estimator receives no outcomes. A fixed attack uses independently selected regions. An adaptive attack selects the next probe to maximize one-step mutual information under its current belief and the known gate. This greedy attack is a reproducible adversary, not a proof of optimal sequential inference or task selection.

The principal analytical model assumes that client and attacker use the same correct mobility and observation models. The original replay study uses a shared approximate model, whereas later stress tests give the attacker more accurate generating probabilities and the post-hoc GeoLife replay uses a stronger population prior and validation-derived mobility estimate. Auxiliary information about an individual’s home, routines, network attachment, or other accounts is outside the main model.

4.2. Branch-Wise Privacy and Maximality

Proposition 1 (Maximal scalar gate). *If $m > 0$ and $\max_i b_i h_i / m > c$, the only feasible gate is $q^* = 0$. Otherwise, Eq. (4) is the largest scalar $q \in [0, 1]$ for which every positive-probability report and silence posterior has peak at most c . When $m = 0$, the output is uninformative and $q^* = 1$.*

Proof. For every $q > 0$ with $m > 0$, Eq. (2) is independent of q . If its peak exceeds c , no positive gate can satisfy the report constraint. Otherwise only the silence constraints remain. For a nonzero silence probability, Eq. (3) is at most c exactly

when

$$q(cm - b_i h_i) \leq c - b_i \quad \text{for all } i. \quad (8)$$

The right side is nonnegative. A nonpositive coefficient on the left imposes no upper bound on q ; a positive coefficient gives the bound in Eq. (4). Their intersection with $[0, 1]$ yields the stated maximum. If silence has zero probability, the corresponding constraint is vacuous and the same expression admits $q = 1$. Finally, $q = 0$ is always feasible because the resulting posterior equals b . $\square$

4.3. Sequential Privacy Guarantee

Corollary 1 (Conditional sequential cap). *Suppose the initial belief has peak at most ρ , the declared mobility and observation models are correct, all public signals are represented by Eq. (1), and every location transition is doubly stochastic. Then BSP maintains posterior peak at most ρ after every output, including under adaptive task selection measurable from public history.*

Proof. A doubly stochastic transition makes each predicted probability a convex combination of previous probabilities, so prediction cannot increase the maximum. Proposition 1 bounds every positive-probability observation branch by ρ . Induction over slots proves the result. Given the public history, the attacker’s adaptive task choice supplies no additional private-state information under the stated measurability assumption. $\square$

A reflecting symmetric grid walk satisfies the transition condition. A general learned human-mobility transition need not. The corollary applies to the attacker’s posterior only when its knowledge and model coincide with the declared model.

4.4. Model-Mismatch Attack and Finite-Model Robustness

Let M_C denote the client’s declared inference model and M_A the attacker’s model. When $M_A \neq M_C$, a gate that is feasible for the client posterior need not

satisfy the attacker’s cap. Incorrect availability, mobility, or prior assumptions can therefore produce an apparently compliant client trace while the attacker’s posterior becomes more concentrated. Section 5 evaluates this failure with an informed attacker.

Proposition 2 (Maximal common finite-model gate). *Within the class of one scalar gate shared by all models, Eq. (5) is the largest $q \in [0, 1]$ for which both positive-probability output branches satisfy $\|b_{t,k}^y(q)\|_\infty \leq c_k$ for every $M_k \in \mathcal{M}$.*

Proof. For each fixed model, Proposition 1 gives the complete feasible scalar set as the interval $[0, q_k^*]$: if the positive branch is infeasible then $q_k^* = 0$; otherwise the silence inequalities supply only upper bounds on q . Requiring all models simultaneously intersects these intervals, giving $[0, \min_k q_k^*]$. Its right endpoint is therefore the maximal common gate. $\square$

Consequently, if the attacker’s exact Bayesian model is in $\mathcal{M}$, its branch constraint is enforced as one member of the intersection; no such conclusion follows for an attacker outside the set.

4.5. Disclosure-Floor Guarantee

Proposition 3 (Maximal asymmetric disclosure-floor-aware scalar gate). *For fixed $(b, S, \alpha, \rho, \tau)$ with $\tau \geq \rho$, DF-BSP returns the largest scalar $q \in [0, 1]$ satisfying $\|b^+\|_\infty \leq c_{\text{DF}}^+$ for every positive-probability report branch and $\|b^-\|_\infty \leq c_{\text{DF}}^-$ for every positive-probability silence branch. When $\tau = \rho$, DF-BSP reduces exactly to BSP.*

Proof. For every $q > 0$, the report posterior is independent of q and has peak $\phi(b, S)$. If this exceeds c_{DF}^+ , no positive gate is feasible. Otherwise the positive constraint is satisfied for every $q > 0$. The remaining silence inequalities are exactly those of

Proposition 1 with $c = c_{\text{DF}}^-$, so their intersection with $[0, 1]$ is $[0, q^*]$ and the closed-form BSP bound remains maximal. Setting $\tau = \rho$ makes $c_{\text{DF}}^+ = c_{\text{DF}}^- = \max\{\rho, \|b\|_\infty\}$ and recovers BSP. $\square$

The result also yields an impossibility within the truthful scalar-gate family: reducing a positive participation probability cannot lower the conditional posterior peak once a successful region report is observed. Fine-region service therefore requires an explicit report-side relaxation such as τ or a different output semantics.

4.6. Scope of the Guarantees

Equation (2) explains why both report and silence must be checked. Scalar thinning changes the probability that a successful report occurs but not its conditional posterior. A single-cell truthful report therefore identifies that cell whenever it is released. Conversely, a report-safe task can still make silence informative because the absence of a report reweights inside and outside states differently. BSP thins only through the public-history scalar gate and sends no separate refusal message.

The same support property rules out interpreting BSP as pure location differential privacy: for a proper region and any nonzero truthful reporting probability, the report event has positive probability inside the region and zero probability outside. Other privacy definitions or mechanisms with altered report semantics remain possible, but they are outside the present guarantee.

A worked report/silence example and the original workflow schematic are provided in the Supplementary Material.

The robust guarantee is finite-model, and all guarantees here are channel-specific. If the attacker’s exact model belongs to $\mathcal{M}$, its constraint is enforced by R-BSP; an attacker outside $\mathcal{M}$ is not covered. Likewise, $\tau > \rho$ explicitly relaxes only the truthful-report branch. It is not evidence that the original ρ cap remains valid after a successful report.

5. Experimental Evaluation

5.1. Experimental Setup

We use three synthetic mobility settings (static, reflecting random walk, and periodic commute) and Microsoft GeoLife trajectories [19, 20, 21]. GeoLife pre-processing scans files in deterministic order and selects the first 48 consecutive in-bounds one-minute bins for each eligible user, without interpolating missing minutes. The rule yields 107 eligible user windows before holdout isolation, split deterministically into 23 development, 17 validation, and 67 test candidates. Because the fitted and replay models consume only the complete 48-slot 8×8 state sequence, effective splits are formed by grouping identical model-input paths and assigning each group to the earliest split in development→validation→test order. This leaves 23 development, 12 validation, and 53 test windows; no complete model-input path crosses the effective splits. Exact GPS-coordinate hashes are retained as a separate provenance audit. Test windows are not used to construct the later data-driven ambiguity set. The 8×8 Beijing grid is intentionally coarse; the study evaluates cell inference, not address recovery.

The public task library contains 152 rectangular regions. Normal tasks are exogenous; on probe slots, the attacker chooses the region with maximal one-step mutual information from the same public library. Delivery, willingness, and timely completion probabilities are 0.9, 0.8, 0.9, giving $\alpha = 0.648$. Normal tasks and exogenous random draws are shared across policies, while adaptive probes are recomputed because the public history changes.

The main study includes no filtering, random and periodic participation, coarse-region controls, BSP, a success-only ablation, and a per-output KL adaptation of Bayesian privacy [17]. We also implement task-channel specializations of PML [4] and PRIVIC [3]. PML-T optimizes a truthful binary channel under the published

pointwise leakage constraints. PRIVIC-T fits its noisy-location channel with the published BA/IBU/GIBU updates and then maps it to truthful task completion; this adapter does not inherit the original geo-indistinguishability guarantee. Full equations, parameter grids, author-code checks, and optimization validation are in the Supplementary Material.

5.2. Data-Driven Ambiguity Set

The original hand-grid R-BSP stress deliberately spans broad (v, α) misspecification, but it is post-hoc. For replay we construct $\mathcal{M}$ using only development and validation users. A coarsened reflecting-walk likelihood uses

$$\Pr(\text{cell change} \mid i) = v d_i / 4,$$

where d_i is the number of in-grid neighbors. The likelihood uses only change versus no-change events; destination direction and jump distance are not fitted. Two thousand cluster-bootstrap resamples of development state-path groups generate candidate v values at the 2.5, 25, 50, 75, and 97.5 percentiles plus the development MLE. Each candidate is paired with a uniform or development-only smoothed population prior. On 2,000 cluster-bootstrap resamples of validation state-path groups, we retain the smallest set covering at least 95% of predictive-likelihood winners. This yields six population-prior models. The development MLE is 0.0514 (95% bootstrap interval 0.0298–0.0754); the validation point estimate is 0.0817. GeoLife has no real task-delivery, willingness, or deadline logs, so $\alpha = 0.648$ remains an explicit protocol parameter rather than a learned quantity.

5.3. Baselines, Metrics, and Inference Units

Privacy metrics are MAP cell hit rate, posterior peak, log loss, calibration, and cap-violation rates. Standard service utility is completed legitimate tasks divided by

externally completable legitimate opportunities. To expose fine-region service, we additionally report retention for 1–8, 9–31, and 32–64 cell regions and inverse-area-weighted retention

$$U_{\text{area}} = \frac{\sum_{t \in \mathcal{L}} Y_t / |S_t|}{\sum_{t \in \mathcal{L}} \mathbf{1}\{X_t \in S_t\} D_t W_t L_t / |S_t|}. \quad (9)$$

Synthetic intervals resample seed clusters and GeoLife intervals resample complete state-path groups; slots and duplicate model-input paths within a group are not treated as independent replicates. Unless stated otherwise, percentile bootstrap intervals use 1,000 resamples. The data-driven and disclosure-floor experiments are post-hoc extensions because the replay test users had been inspected in earlier analyses; they are not presented as independent confirmatory validation.

5.4. Effectiveness of BSP and Recent Comparators

Under the original uniform-prior attacker, BSP reduces declared-model posterior concentration while retaining nonzero service, but completed tasks are dominated by large regions. A development-only no-output modal-cell predictor already attains 16.76% hit rate on isolated test positions, showing that a uniform-prior attack is not a deployment upper bound. This motivates the stronger population-prior replay attacker used below.

At opportunity retention 0.5 in the static scenario, exploratory envelope interpolation gives BSP MAP hit rate about 8.0%, versus 16.7% for PML-T and 18.6% for PRIVIC-T; the paired differences are approximately -8.8 and -10.6 percentage points. Walk, commute, and replay comparisons do not show a consistent advantage over both recent comparators, and some matched-retention intervals cross zero. Because the policy envelopes are selected from evaluation results, these are exploratory comparisons rather than independently tuned superiority claims. Complete curves, calibration, ablations, and interpolation diagnostics are in the Supplementary Material.

5.5. *Model-Mismatch Attack*

Nominal BSP is fragile when the attacker knows a better model. In the informed synthetic stress at $\rho = 0.1$, local-cap violations reach 12.6% when the client underestimates availability, 2.6% when it overestimates availability, 2.8% when it underestimates movement, and 14.6% when it overestimates movement. These failures motivate the robust extension rather than contradicting its model-conditional theorem.

5.6. *Finite-Model Robustness*

A nine-model hand-grid R-BSP stress contains the informed attacker’s exact $(v, \alpha) = (0.3, 0.648)$ model. At $\rho = 0.1$, it records zero local-cap violations in the four misspecified conditions because the informed attacker’s exact generating model is included in the set, but it retains only about 39.7% of externally completable normal-task opportunities. This run validates implementation and exposes robustness cost; it is not evidence of generalization beyond the set.

A separate boundary run fixes the same model set and changes the correct attacker model. An outside-set availability value $\alpha = 0.98$ restores a 6.93% local-cap violation rate (95% cluster-bootstrap interval 5.70–8.31%). Interior unlisted values produce zero or rare violations in this finite sample. The guarantee is therefore explicitly finite-model. Full stress tables and boundary conditions are in the Supplementary Material.

5.7. *Data-Driven Robustness*

Table 1 first shows that a defensible model-set construction does not remove the strict-cap utility problem. Against the stronger replay attacker described above, nominal BSP exceeds the attacker’s own local cap on 24.5% of slots and the earlier hand-grid R-BSP on 19.1%. The validation-best single-model BSP reduces the

Table 1: Post-hoc GeoLife data-driven robustness at $\rho = 0.1$. The replay attacker uses the development-only smoothed population prior and validation point mobility estimate. “Local viol.” tests the attacker’s pre-output local cap. Weighted retention is Eq. (9); small retention covers 1–8-cell opportunities.

Policy	MAP hit	Local viol.	Ret.	Weighted ret.	Small ret.
Uniform-prior BSP	33.8%	24.5%	43.6%	10.4%	0.0%
Validation-best single BSP	12.1%	0.1%	30.7%	5.8%	0.0%
Hand-grid R-BSP	23.4%	19.1%	44.1%	10.1%	0.0%
Data-driven R-BSP	12.9%	0.2%	18.8%	3.0%	0.0%

local-cap violation rate to 0.1% but retains only 30.7% of opportunities, showing that improved model choice explains part, but not all, of the robustness gain. The six-model data-driven R-BSP records 0.2% local-cap violations in this run and a 12.9% MAP hit rate, but retains only 18.8% of opportunities and 3.0% inverse-area-weighted utility. All four strict policies retain zero 1–8-cell opportunities. The data-driven set is therefore better grounded than the hand grid but substantially more conservative; model-set provenance and service viability are separate problems.

5.8. Fine-Grained Service under Disclosure Floors

Table 2 quantifies the service alternative. On 20 new synthetic walk seeds, ordinary BSP retains no small-region opportunities and has weighted retention 12.3%. Asymmetric DF-BSP with $\tau = 0.5$ raises small-region retention to 56.4% (cluster-bootstrap 95% interval 41.7–69.8%) and weighted retention to 43.1%, with no observed violation of its declared branch-specific caps. The silence branch continues to use the original BSP cap. The original BSP local cap is violated on 8.4% of slots because the report-side relaxation can increase posterior concentration beyond the strict BSP branch bound; MAP hit rate rises from 4.9% under BSP to 10.8%.

The stronger replay experiment shows the same trade-off under prior and mo-

Table 2: Disclosure-floor-aware service frontier at $\rho = 0.1$. “BSP viol.” reports violation of the original BSP local cap $\max\{\rho, \|b\|_\infty\}$; “Branch viol.” reports violation of the declared asymmetric caps in Eq. (7). Weighted and small-region retention emphasize fine-grained service. RDF-BSP uses the six validation-selected models.

Scenario	Policy	τ	BSP local viol.	Branch viol.	Weighted ret.	Small ret.
Walk	DF-BSP	0.125	7.8%	0.0%	14.8%	0.0%
Walk	DF-BSP	0.25	7.7%	0.0%	19.2%	1.8%
Walk	DF-BSP	0.5	8.4%	0.0%	43.1%	56.4%
Walk	DF-BSP	1	8.6%	0.0%	67.0%	67.3%
Walk	BSP	–	0.0%	0.0%	12.3%	0.0%
GeoLife	DF-BSP	0.125	25.7%	25.3%	13.0%	0.0%
GeoLife	RDF-BSP	0.125	9.2%	0.2%	5.9%	0.0%
GeoLife	DF-BSP	0.25	26.5%	20.4%	18.0%	5.9%
GeoLife	RDF-BSP	0.25	16.5%	2.5%	12.1%	0.0%
GeoLife	DF-BSP	0.5	28.8%	16.7%	30.9%	35.3%
GeoLife	RDF-BSP	0.5	14.7%	2.0%	26.0%	23.5%
GeoLife	DF-BSP	1	38.0%	24.1%	44.5%	44.1%
GeoLife	RDF-BSP	1	15.7%	3.1%	46.3%	41.2%
GeoLife	Uniform BSP	–	24.5%	24.5%	10.4%	0.0%
GeoLife	Val.-best BSP	–	0.1%	0.1%	5.8%	0.0%

bility mismatch. At $\tau = 0.5$, non-robust DF-BSP retains 35.3% of small-region opportunities and 30.9% weighted utility, but violates its declared branch-specific cap on 16.7% of slots because the stronger replay attacker is outside the client model. Data-driven RDF-BSP lowers that violation rate to 2.00% (cluster-bootstrap 95% interval 1.40–2.64%), with 23.5% small-region retention (95% interval 10.0–39.4%) and 26.0% weighted utility. Its original BSP local cap is violated on 14.7% of slots. At $\tau = 1$, the report-side threshold is vacuous and should be read as a service control rather than a privacy operating point. No tested τ is declared optimal.

5.9. *Efficiency*

For the 64-cell main grid, the mean of per-trajectory median BSP gate times is approximately 16.8–16.9 microseconds; the mean of per-trajectory 95th percentiles is 18.2–18.7 microseconds. The iterative KL implementation has median summaries of about 326–341 microseconds. These are local Python gate timings, excluding probe search, posterior metrics, loading, and communication. Methods ran in separate batches, so the numbers are descriptive implementation costs, not controlled cross-device speedups. Smartphone latency, energy consumption, and cryptographic execution time were not measured. In the recent extension, condition means of trajectory-median gate times range from 12.1 to 14.0 microseconds for PML-T and 2.0 to 3.0 microseconds for PRIVIC-T. PRIVIC training across all 24 channels takes 182.3 seconds separately; online timings exclude that offline fitting. These batch-specific measurements do not establish a controlled speedup.

5.10. *Discussion and Limitations*

The supported positive finding is narrow: a public-history participation rule can reduce the declared model’s inference accuracy while retaining some large-region task opportunities. Checking both possible outputs before participation gives a clear model-conditional bound. Static comparisons provide the clearest recent-baseline evidence. Coarse replay improvements are specific to the uniform-prior inference model and do not bound practical observer accuracy; walk and periodic comparisons show no consistent advantage over both recent comparators.

First, nominal BSP relies on one declared belief model. R-BSP turns this limitation into an explicit finite ambiguity-set guarantee by intersecting member-wise constraints, and the hand-grid stress test shows the expected removal of local-cap violations when the informed attacker’s exact model is included. The later development/validation construction replaces the hand-grid choice for replay mobility and

prior uncertainty with a reproducible selection rule. The smallest set meeting the 95% target nevertheless contains all six population-prior mobility candidates and covers 100% of validation-bootstrap winners. Strict data-driven R-BSP then retains only 18.8% of isolated replay opportunities. Thus “data-driven” does not mean “low cost.” The boundary experiment also shows that an outside-set availability model at $\alpha = 0.98$ can restore violations. Moreover, α itself is not learned because the trajectory dataset contains no task-availability observations. A deployment would need task-process data, distribution-shift handling, and prospective evaluation of the set-construction rule.

Second, the model omits informative measurement contents, acknowledgments, payment records, network observations, and within-slot timing. Common-deadline release is a protocol requirement, not an implemented property of a real platform. General collection windows need a state describing visits or acquisition time rather than only the current sampled cell. The on-time probability sweep does not replace such a model.

Third, the data are short, selected, historical, and coarse. One 48-minute window per user does not describe a full day; excluding users without consecutive samples can change the mobility mix. 5 isolated test windows have no observed cell change, and movements within a cell of roughly 4.2 km by 4.3 km at the study latitude are invisible at this resolution. Tasks are simulated on public movement traces. The experiments estimate neither the prevalence of current attacks nor the probability of identifying a home. The development-fitted no-output diagnostic quantifies one weakness of the uniform prior. The later post-hoc extension does rerun adaptive probes with a development-only smoothed population prior and validation-derived mobility estimate, but because the replay test users had already been inspected in earlier analyses, those results are exploratory rather than an independent confirmation.

Fourth, the attack is greedy with a fixed probe budget. Longer-horizon task design, coordinated accounts, location-dependent availability, and personalized prior knowledge may change the frontier. Strict truthful reporting has a structural granularity cost: scalar thinning cannot reduce the conditional leakage of a success. DF-BSP makes the resulting service relaxation explicit through τ and improves fine-region retention while keeping the silence branch at the original BSP cap; it changes only the truthful-report privacy contract whenever $\tau > \rho$. The inverse-area utility is deliberately diagnostic rather than an economic welfare model, and the tested τ values do not identify an optimal operating point. A real deployment would need application-specific task values or measurement reconstruction quality and a policy for choosing τ . Independent account repetitions do not constitute a multi-user allocation or platform-capacity benchmark.

Finally, comparisons depend on finite parameter grids and exploratory interpolation. Bootstrap intervals condition on the selected traces, tasks, and clustering, without resolving distribution shift or multiple testing. The KL and recent-method extensions were added after the main run. Recent policy envelopes are selected on the evaluation set; their bootstrap intervals describe this exploratory selection procedure rather than validating an independently tuned policy. PRIVIC uses one noisy training seed and some IBU/GIBU loops reach their iteration limits; small likelihood gaps do not establish distribution recovery. Its true-eligibility task adapter changes the published output semantics. The full-text comparison with HECTA establishes differences in observations, trust assumptions, and privacy objectives, but does not provide a head-to-head system evaluation or an exhaustive novelty assessment.

5.11. Reproducibility and Research Record

The local artifact contains configurations, deterministic preprocessing, user splits, hashes, seeds, compressed event logs, trajectory-level rows, analysis code, generated figures, and LaTeX sources. The public-observation module receives no ground truth or future positions. Evaluation logs retain simulator truth for auditing and are not the attack interface. Plots and tabulated numbers are generated from saved results. The additional prior diagnostic, task-area table, pooled raw completions, and expanded comparison plot are regenerated by the review-diagnostics script. Its input hashes, per-user values, and per-task area records are saved separately from the original simulations.

The automated suite contains 30 tests. Core mechanism tests cover posterior branches, nominal/robust/disclosure-floor gate bounds and reductions, an informed-attacker R-BSP integration case, the coarsened mobility estimator, KL feasibility, transition conservation, and boundary cases. Pipeline and comparator tests cover duplicate or expired tasks, missing minutes, split separation, paired normal tasks, numerical replay, BA/IBU conformance to the PRIVIC author code, PML optimality against a general linear program, vector-valued observation channels, and policy-envelope agreement with a mixture linear program. The corrected pilot, interrupted KL run, and initial shorter PRIVIC training are retained separately and excluded from the final comparisons. Completed runs contain no recorded simulation failures.

6. Conclusion

Sequential regional task outputs can reveal location despite local storage of precise coordinates, and probabilistic silence must be included in that analysis. BSP provides a maximal scalar gate for both posterior branches under explicit model assumptions, while R-BSP extends the construction to a finite model set.

Development/validation-based model selection makes that set more defensible but reveals that strict robust posterior caps can be prohibitively conservative: the isolated replay experiment retains only 18.8% of opportunities. This is linked to a structural truthful-report floor that scalar thinning cannot remove. DF-BSP and RDF-BSP therefore expose a separate report-side threshold τ and recover measurable fine-region service while reporting, rather than hiding, exceedance of the original ρ cap. At the illustrative $\tau = 0.5$, robust replay filtering retains 23.5% of small-region opportunities and 26.0% inverse-area-weighted utility, with a 2.00% observed branch-cap violation rate and original BSP local-cap violations on 14.7% of slots. The contribution is consequently a channel-specific framework for strict and explicitly relaxed participation filtering under finite model uncertainty, not a guarantee against arbitrary auxiliary knowledge or a complete mobile-crowdsensing privacy system.

Data and code availability

Code, configurations, deterministic preprocessing, processed GeoLife split files, simulation outputs, analysis scripts, tests, and manuscript sources supporting the reported results are available in the project repository at <https://github.com/js110/Bpaper>. The original Microsoft GeoLife trajectory archive is obtained separately from its recorded public source and is not redistributed as part of the manuscript package. Third-party article full texts used for literature checking are likewise excluded from the distributable artifact. The repository is version controlled; a versioned release or immutable archival identifier can be associated with the submitted version separately.

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